Review: Similar figures

1 It has no units.

2 Smaller

3 2.5

4

- **a** Yes
- **b** No
- **c** No
- **d** No

5

- **a** 2
- **b** 6

6 $\frac{1}{3}$

7 4

8 $\frac{1}{3}$

9 Reduction factor$=\frac{4}{5}$, Enlargement factor$=\frac{5}{4}$

10 4

11

- **a** Yes
- **b** 3

12 3

13 22.5 cm

14

- **a** 71.3 cm
- **b** 68.2 cm
- **c** 58.9 cm

15

a $19.32\ \text{cm}$

b $23\ \text{cm}$

c $18.4\ \text{cm}$

16

a 27

b 3

17 $4\ \text{cm}$

18 $21\ \text{cm}$

19

a $30\ \text{cm}$

b 60°

20

a 5

b $15^2 + 20^2 = 25^2$

$225 + 400 = 625$

21

a Yes

b No

c No

d No

e No

22

a 3

b $189\ \text{cm}^2$

23

a $2\ \text{cm}$

b $36\ \text{cm}^2$

c 9

24 $270\ \text{cm}^2$

25

a $60\ \text{cm}^2$

b $240\ \text{cm}^2$

c 2

d 4

26

a 2

b $\frac{1}{2}$

c 6 units^2

d 24 units^2

e 4

27 $\triangle ABC \sim \triangle DEF$

28 **B** and **C**

29

a Similar

b $\overline{EF}$

c $\angle EHG$

30

a Yes

b Yes

c Yes

d Yes

31

a Yes

b

i No ii No iii Yes

32 No

33

a Yes

b 6

34

a $\frac{3}{2}$

b 12

35

a $\frac{2}{7}$

b 4

36

a

i 2 ii $x = 20$

b

i 2 ii $x = 40$

c

d

37 2

38

a Yes

b Corresponding angles are not equal.

39

a Yes

b Corresponding sides are not equal.

40

a Yes

b Yes

c No

d No

e Yes

f No

Additional questions

41

a Similar figures

b Not similar figures

c Not similar figures

d Similar figures

e Similar figures

f Similar figures

42

a $\angle IHN$

b $\angle KJI$

43 5.5

44

a

i A and B are similar.

ii A and B are not congruent.

iii A is dilated by the scale factor 2.

b

i A and B are similar.

ii A and B are not congruent.

iii A is dilated by the scale factor $\frac{1}{3}$.

c

i A and B are similar.

ii A and B are congruent.

iii A is dilated by the scale factor 1.

45

- Corresponding sides must be proportional to each other.
- Corresponding angles must be equal.
- One shape can be mapped to the other through a series of transformations.

46 For two similar figures to be congruent, the scale factor of the dilation must be 1.